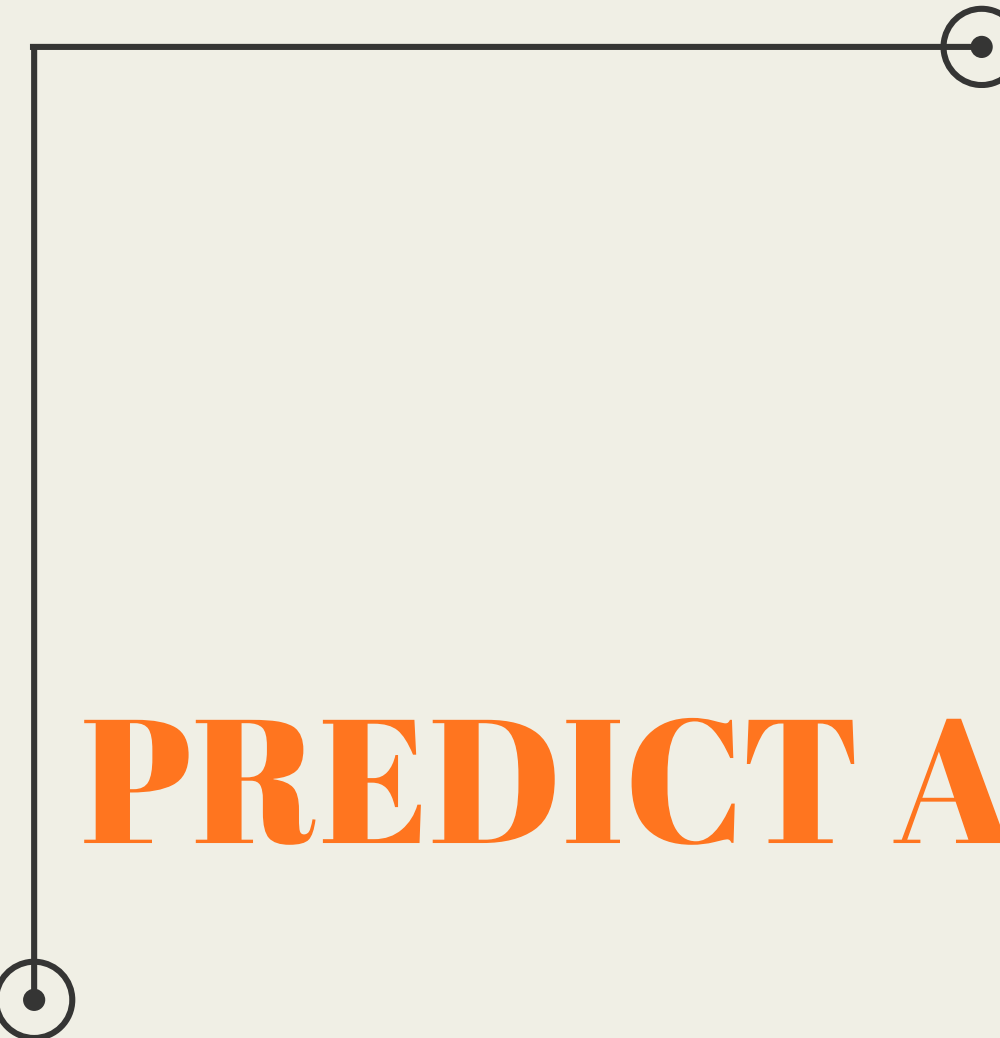
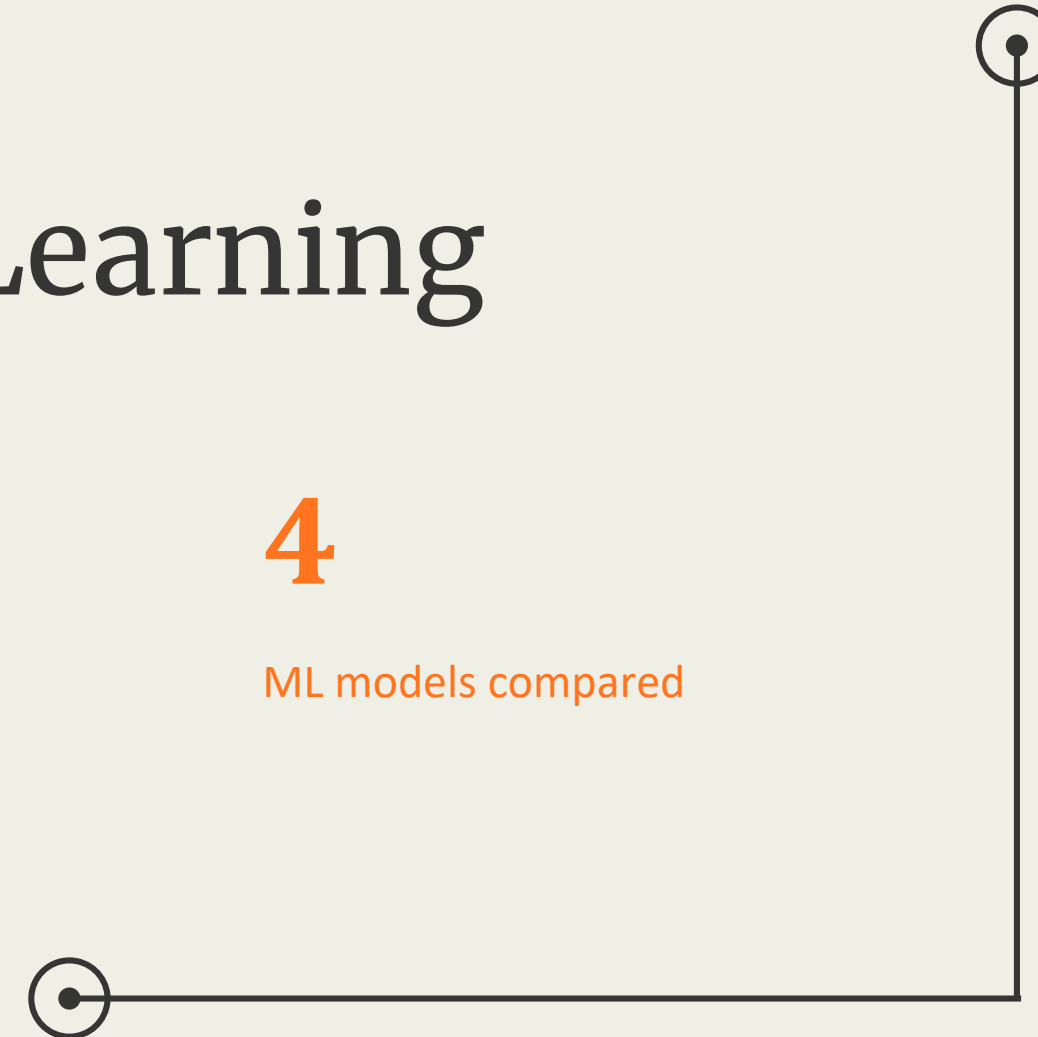




CAN AI

PREDICT A STROKE BEFORE IT HAPPENS

Stroke Prediction Using Machine Learning



5,110

Patient records

12

Clinical features

4

ML models compared

Team Members



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OVERVIEW



Stroke occurs when the blood supply to the brain is interrupted

→ This can cause serious damage, leading to disability or even death

Several Clinical factors such as age, hypertension, heart disease, glucose level

and BMI affect stroke Risk

→ Understanding these factors helps in identifying high-risk patients early

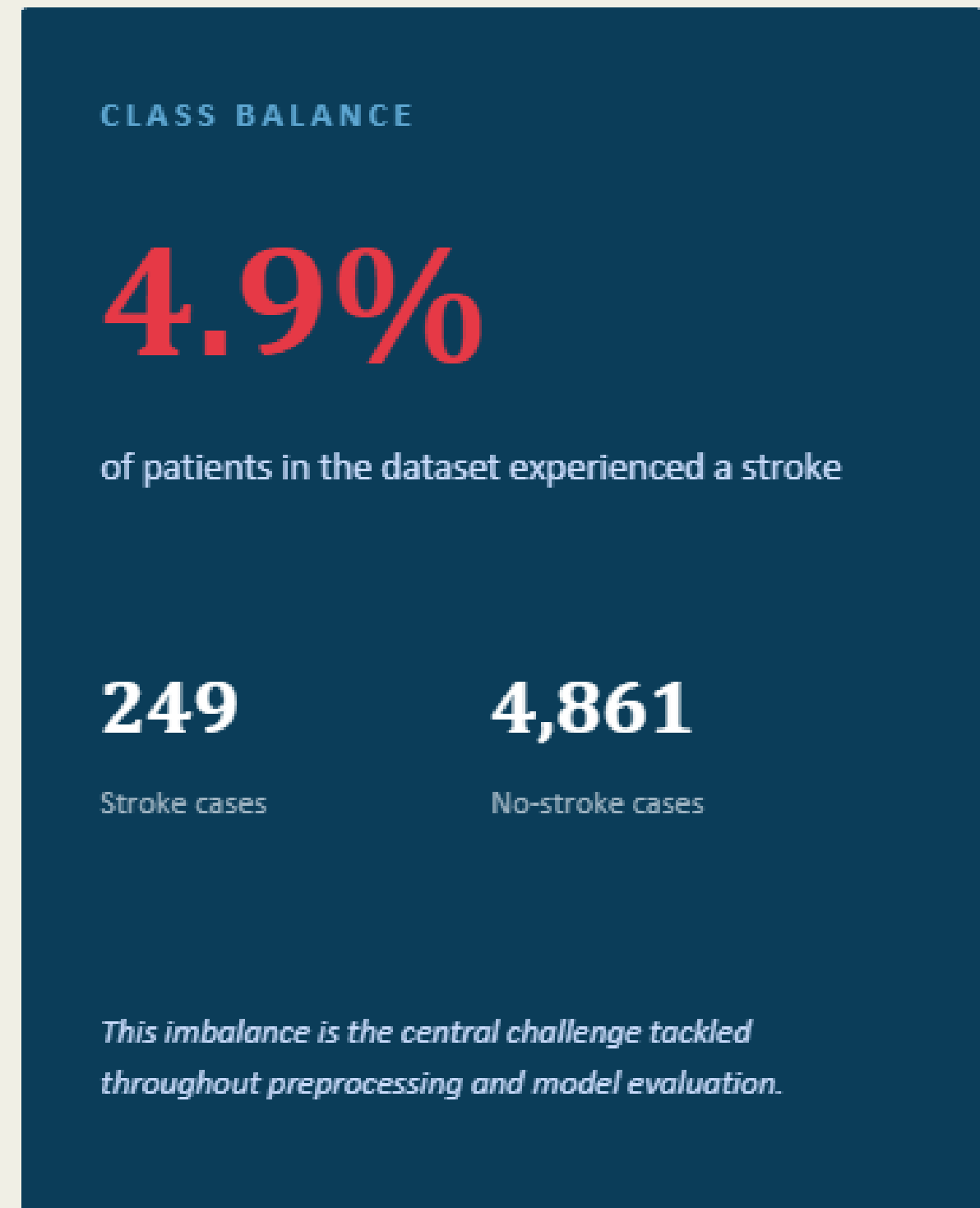
Medical systems collect large amounts of patient health data

→ Analyzing these complex data manually is difficult and time-consuming

Machine Learning algorithms can analyze clinical data and detect hidden patterns

→ This allows the development of models that can predict stroke risk accurately

This project Aims to use Machine Learning techniques to predict stroke occurrence based on patient health information.



PROBLEM STATEMENT



Stroke prediction is challenging because many clinical factors are involved.

→ This makes early identification of high-risk patients difficult.

Traditional methods depend mainly on manual medical analysis.

→ This can be time-consuming and may not discover complex patterns in data.

Large healthcare datasets contain valuable information about stroke risk.

→ However, extracting useful insights from this data requires intelligent analysis methods.

Machine Learning provides an effective approach to analyze patient data.

→ This can help create predictive models for early stroke risk detection.

OBJECTIVES



Collect and analyze clinical data related to stroke risk factors.

→ **To understand the relationship between patient features and stroke occurrence.**

Apply Machine Learning algorithms on patient health data.

→ **To build models capable of predicting stroke risk.**

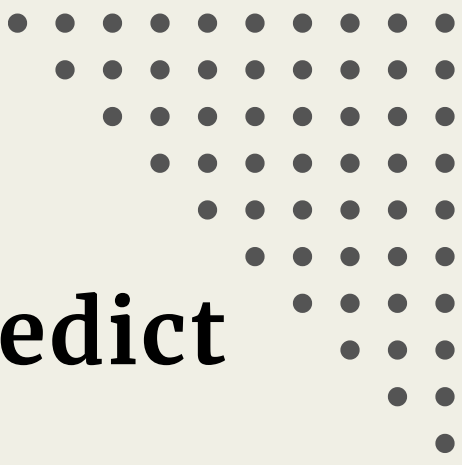
Evaluate and compare different Machine Learning models.

→ **To identify the model with the best prediction performance.**

Develop an intelligent prediction system for early stroke detection.

→ **To support faster decision-making and improve preventive healthcare.**

DATASET DESCRIPTION



The dataset contains clinical information collected from patients to predict stroke occurrence

It provides medical factors that can be used to identify stroke risk.

The Dataset includes 10 Clinical features related to patient health conditions

These features represent different factors that may affect stroke probability.

The Target Variable is "Stroke"

It indicates whether the patient experienced a stroke or not.

The Problem is a Binary Classification Task

The model classifies patients into two categories: Stroke (1) or No Stroke (0).

PATIENT DATA → **MACHINE LEARNING MODEL** → **STROKE / NO STROKE**

DATASET FEATURES & DATA UNDERSTANDING



The dataset contains different clinical and demographic features related to patient health.

These features help the Machine Learning Model understand factors associated with stroke risk.

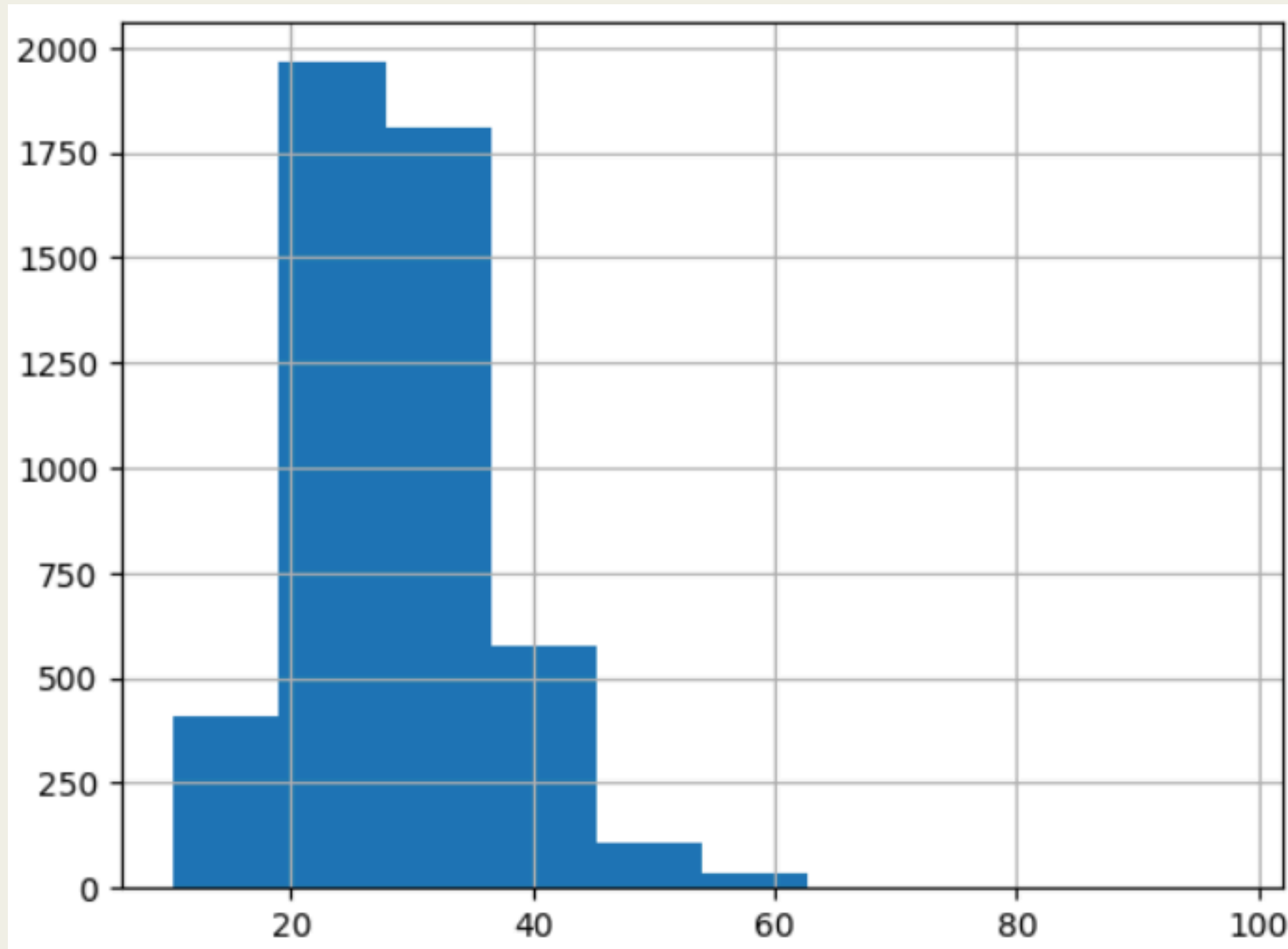
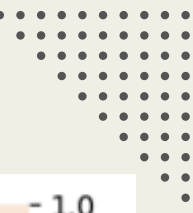
The main features include:

Age → Patient's age \ Gender → Patient's gender. \ Hypertension → Whether the patient has high blood pressure.
Heart Disease → Whether the patient has heart disease. / Average Glucose Level → Average blood measurement.
BMI → Body Mass Index. / Smoking Status → Patient's smoking condition. / Work Type → Type of employment.
Residence Type → Urban or rural residence. / Marital Status → Patient's marital status.

The target feature is Stroke.

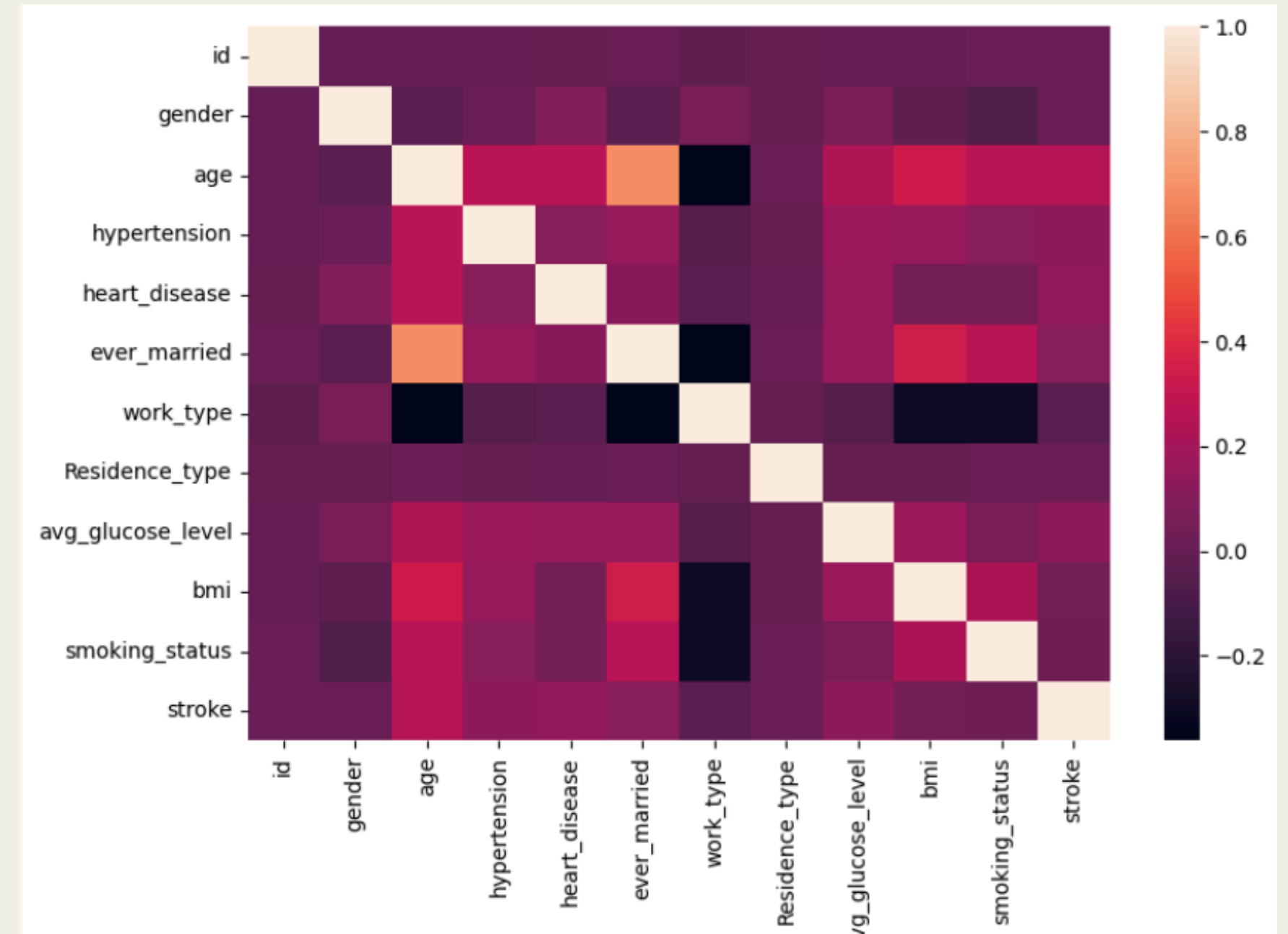
→ It Represents the final prediction outcome: Stroke or No Stroke.

DATASET FEATURES & DATA UNDERSTANDING



BMI Distribution Analysis

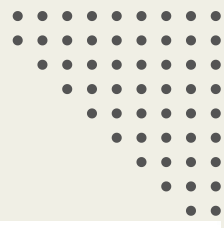
Shows the distribution of BMI values among .



Correlation Heatmap Analysis

Shows the relationships between patient features in the Stroke dataset.

DATA PREPROCESSING



Raw medical data cannot be directly used by Machine Learning models.

Therefore, data preprocessing is required to prepare the dataset. Missing values are identified and handled.

This improves data quality and prevents errors during training. Categorical features are converted into numerical values using encoding techniques.

This allows Machine Learning algorithms to process the data. Features are scaled when needed.

This ensures that all features have a similar influence on the model.

The dataset is divided into training and testing sets.

The model learns from training data and is evaluated on unseen testing data.

Raw Dataset → Data Cleaning → Encoding → Feature Scaling → Train Test Split → Train → Machine Learning Model

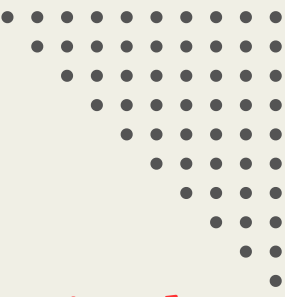
BEFORE → AFTER ENCODING

Column	Raw value	Encoded
gender	Male / Female	1 / 0
ever_married	Yes / No	1 / 0
work_type	Private, Self-employed...	0–4
Residence_type	Urban / Rural	1 / 0
smoking_status	smokes, never...	0–3

Missing BMI values (201) replaced with the median of 28.1

Median chosen over mean because the BMI distribution is skewed with outliers on the high end.

EXPLORATORY DATA ANALYSIS (EDA)



Dataset Overview :

Understanding the dataset structure, size, and feature types. / Identifying numerical and categorical variables.

Data Quality Analysis :

Checking missing values and duplicate records. / Ensuring data consistency before training.

Feature Analysis :

Studying the distribution of clinical features. / Understanding how patient characteristics relate to stroke occurrence.

Statistical & Visual Analysis :

Using statistical measurements and visualization techniques. / Discovering patterns, correlations, and unusual values.

Class Distribution Analysis :

Examining the balance between Stroke and No Stroke cases. / Understanding its impact on model training.

EDA Insights :

Selecting important features. / Improving preprocessing decisions.

Preparing better input data for Machine Learning models.

MODEL TRAINING & EVALUATION



- **Teaching the Models**
 - The dataset is divided into training and testing sets, allowing the models to learn meaningful patterns from patient clinical data.
- **Testing the Models**
 - The trained models are evaluated using unseen data to examine their ability to make accurate stroke predictions.
- **Measuring the Performance**
 - Performance metrics such as Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix are used to assess prediction quality.
- **Comparing the Results**
 - The evaluation results are analyzed to compare the strengths and weaknesses of each Machine Learning model.
- **Choosing the Best Model**
 - The model with the most reliable and consistent performance is selected for stroke prediction.

RESULTS & MODEL COMPARISON

- Observe the Results

Analyze the performance of each Machine Learning model using the evaluation metrics.

- Compare the Models

Compare the strengths and limitations of the models based on their prediction performance.

- Make the Decision

Determine the most suitable model for supporting early stroke risk prediction.

Best Model : Random Forest

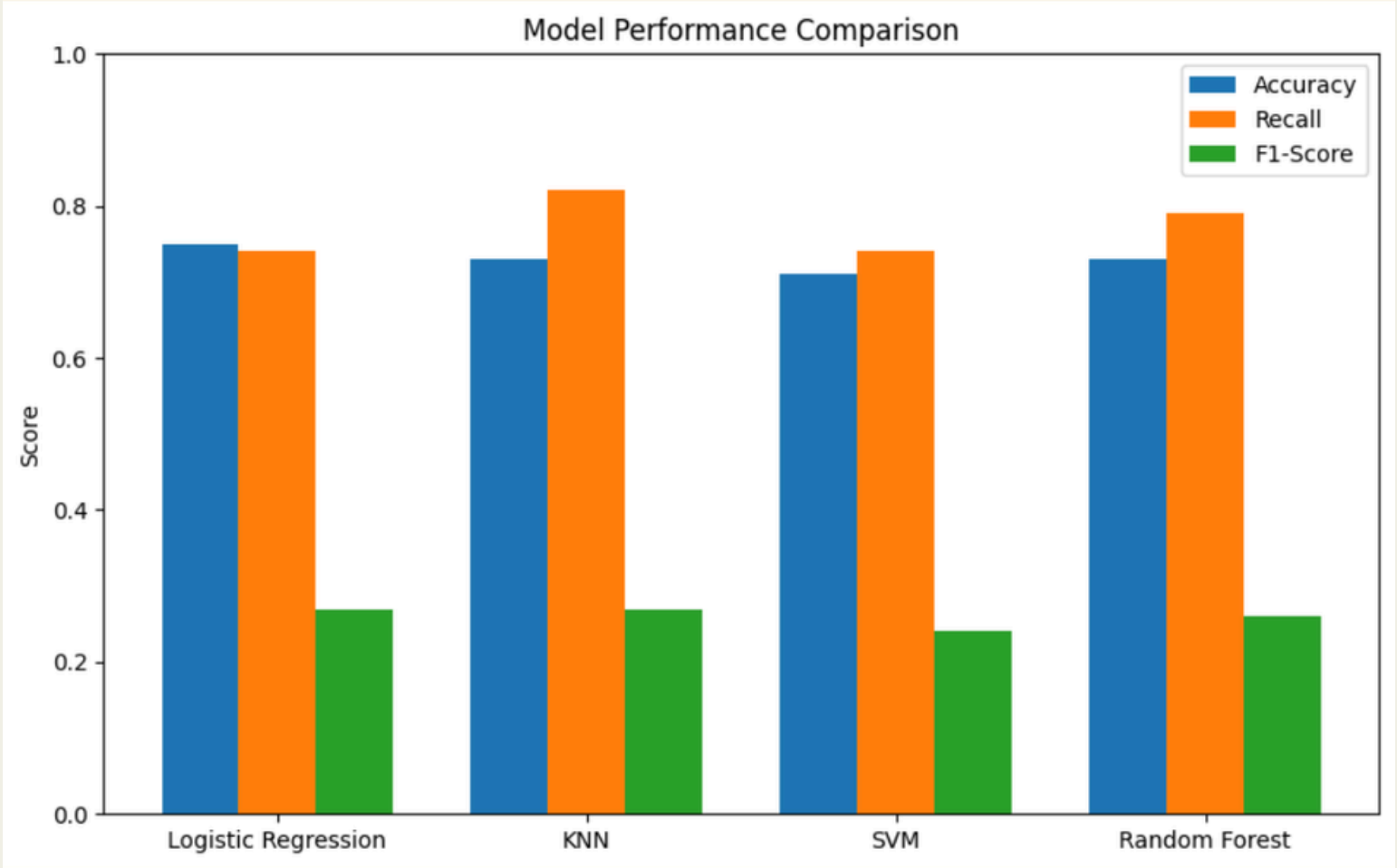
- Achieved high Recall (81%), meaning it detected most stroke cases.
Missed only 12 out of 62 stroke cases (FN = 12).

- Provides Feature Importance, which helps explain the main factors affecting stroke prediction.

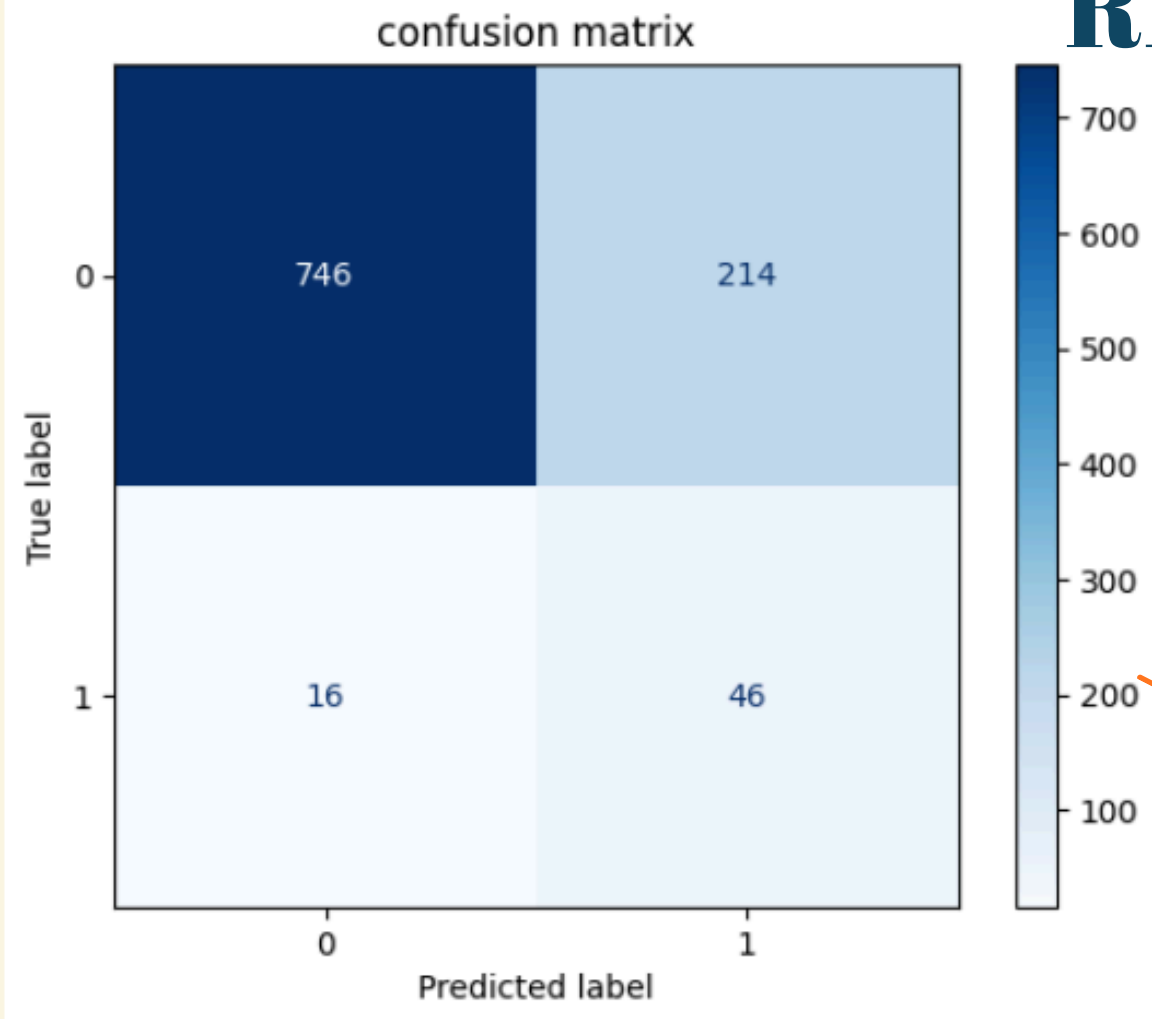
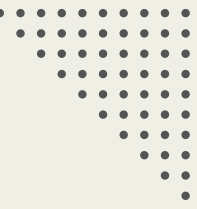
Why NOT Logistic Regression?

Although Logistic Regression achieved higher accuracy (77%), it had lower Recall (74%) and missed more stroke cases (FN = 16)
Random Forest was selected because detecting stroke cases is more important than overall accuracy.

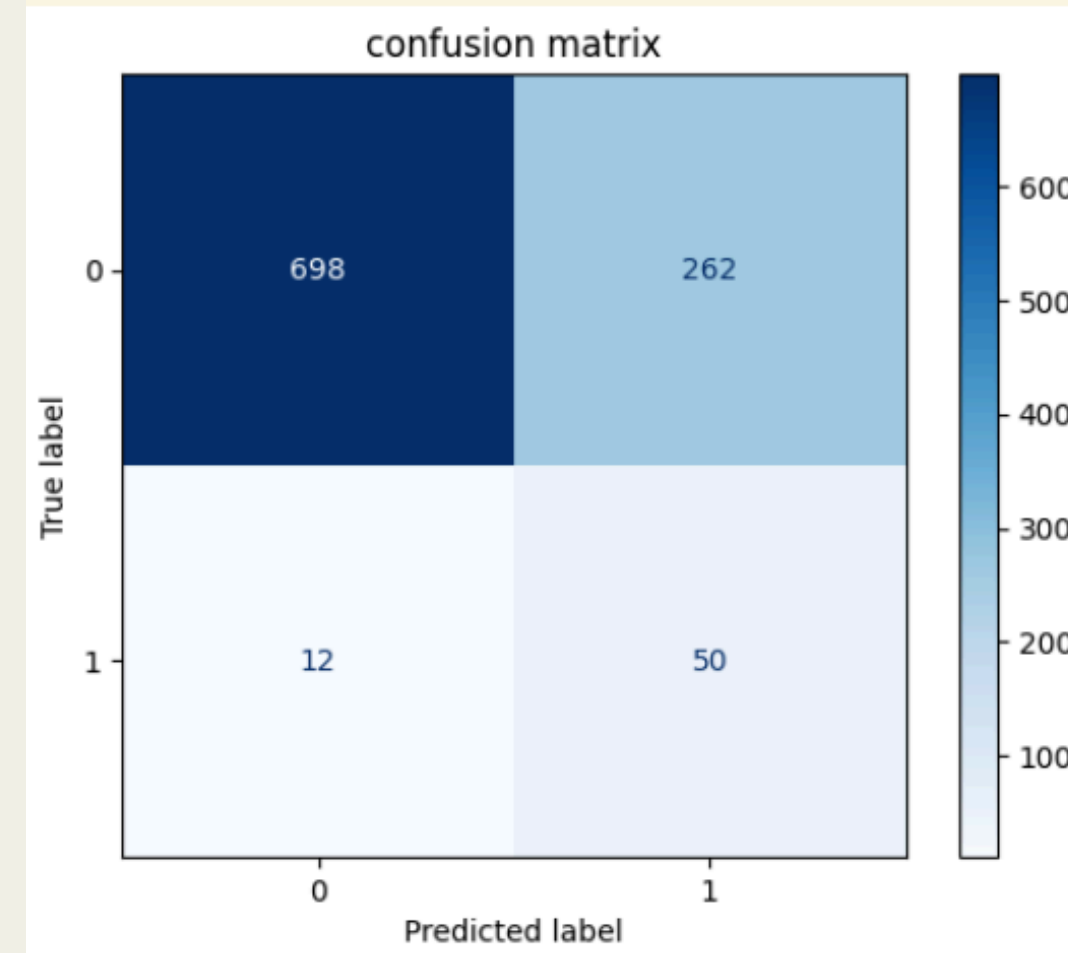
Model	Accuracy	Precision	Recall	F1-Score	Roc-AUC
Logistic Regression	77%	18%	74%	29%	75.95%
KNN	73%	16%	81%	27%	76.68%
SVM	78%	16%	61%	26%	70.33%
Random Forest	73%	16%	81%	26%	76.31%



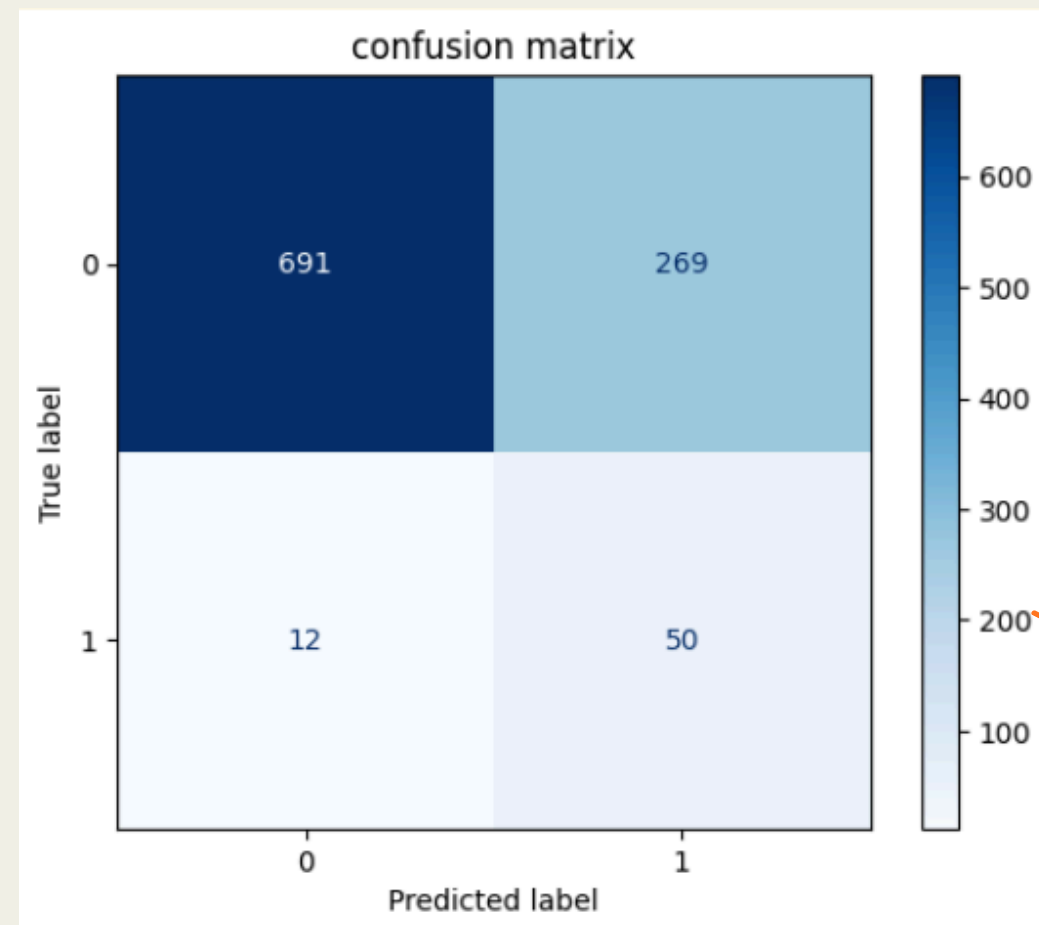
RESULTS & MODEL COMPARISON



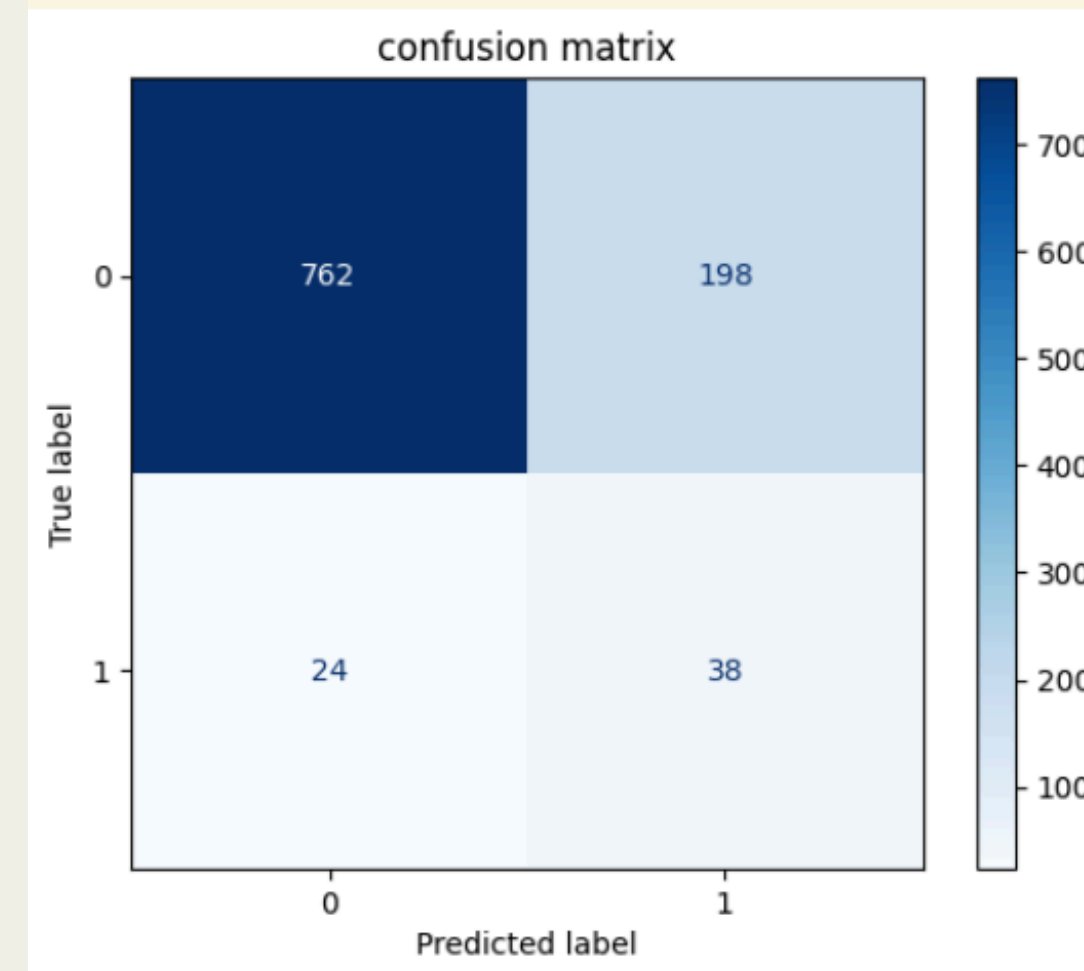
Confusion Matrix Evaluation
LOGISTIC REGRESSION



Confusion Matrix Evaluation
SVM

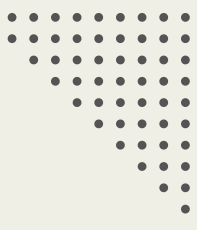


Confusion Matrix Evaluation
Random forest



Confusion Matrix Evaluation
KNN

FEATURE IMPORTANCE ANALYSIS



Why does Random Forest make this prediction?

- Random Forest analyzes the contribution of each feature to identify the factors that have the highest impact on stroke prediction.

Most Important Features:

1. Age : Older age increases the probability of stroke.
2. Average Glucose Level : Higher glucose levels are associated with higher stroke risk.
3. BMI : Body weight indicators can affect stroke probability.
4. Hypertension : High blood pressure is a major stroke risk factor.
5. Heart Disease : Existing heart conditions increase stroke risk.

Testing Random Forest on a New Patient

The trained model was used to predict the stroke risk for a new person's data.

Patient Information:

1. Age: 67 years
2. Hypertension: Yes
3. Heart Disease: Yes
4. Average Glucose Level: 180 mg/dL
5. BMI: 32
5. Married : Yes
6. Smoke : Yes

The model identifies high-risk patterns based on patient health characteristics and provides an early prediction that can support medical decisions.

Risk assessment

Two models, one dataset (5,110 patient records) — read side by side.

LOGISTIC REGRESSION

94.3%

Elevated risk

RANDOM FOREST

79.0%

Elevated risk

Elevated risk — models diverge somewhat



Average estimated probability 86.6%. The Random Forest and Logistic Regression differ by about 15 points here — worth reading both scores rather than one alone.

WHAT THE RANDOM FOREST WEIGHS MOST

Age		46.3%
BMI		20.4%
Glucose level		19.5%
Smoking status		5.4%
Ever married		3.0%
Hypertension		2.9%
Heart disease		2.6%



CONCLUSION & FUTURE WORK



- **How Machine Learning Helps Us :**

Machine Learning helps analyze large amounts of medical data, discover hidden patterns, and support early stroke risk prediction, which can assist in faster decision-making.

- **Machine Learning for Stroke Prediction :**

Machine Learning models can use patient health information to identify patterns related to stroke risk and provide accurate predictions.

- **Best Model Performance :**

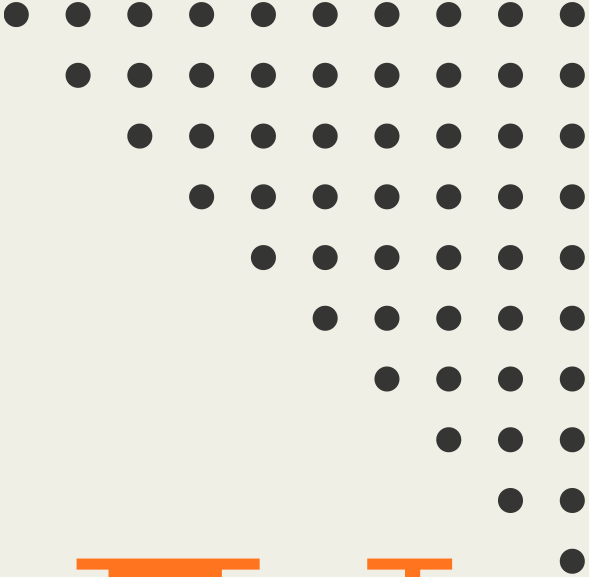
Random Forest was selected as the best model because it achieved high Recall (81%), allowing it to detect more stroke cases and reduce missed predictions.

- **Important Risk Factors :**

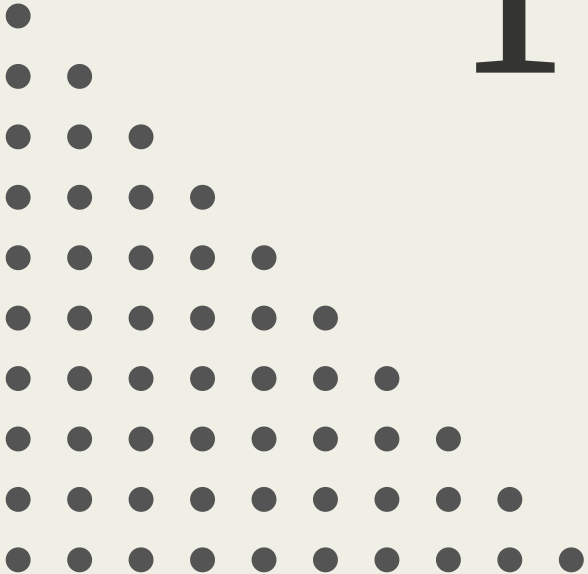
Feature Importance showed that factors such as Age, Glucose Level, BMI, Hypertension, and Heart Disease have a strong impact on stroke prediction.

- **Future Improvements :**

Future work can include using larger datasets, improving data balance, and developing real-time healthcare systems for early stroke risk assessment.



THANK YOU



THE **OUTSHINEING** TEAM

